

University of Virginia
DS 5110 Big Data Systems
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Final Project Instructions
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Instructions

Students will form teams of roughly 3-4 students with the goal of completing a substantial course project. This document outlines the necessary components of the final project.

Deliverables

The project has four deliverables:

1. Project Proposal (5% of overall grade)
2. Code (5% of overall grade)
3. Presentation (10% of overall grade)
4. Paper (10% of overall grade)

These are detailed below, along with general components of the project.

Components

1. Project Proposal

The project proposal will outline the project idea and plan. Since the project will be done over several months, it should be substantive and relevant. The proposal can be as short as two paragraphs and as long as two pages.

It will be submitted part-way into the course (the due date will be announced and posted on Canvas). It should include this information at a minimum:

i. Research aim

This is the motivation for the project. Examples:

- a. We will predict which businesses will go bankrupt within 12 months
- b. We will construct a model to rank-order the risk of hospital patients with a given infection

ii. A plan for the data to be used

This should include the source and a brief summary.

Note about Data

There are several free sources of online data including the ones below. If you need suggestions, I'm happy to discuss. Larger datasets with more opportunity to show your data ninja skills are better. Text

data provides the opportunity to apply NLP. This is an opportunity to be creative and show off! ***Please limit the size of the dataset to 50GB.*** This will minimize waiting time while iterating.

Physionet

Physiological signals including ECGs

<https://physionet.org/>

NYC OpenData

<https://opendata.cityofnewyork.us/>

Kaggle

<https://www.kaggle.com>

StatLib---Datasets Archive

<http://lib.stat.cmu.edu/datasets/>

Large Global Address Dataset (language classification, for example)

<https://openaddresses.io/>

UCI Machine Learning Repository

<http://archive.ics.uci.edu/ml/index.php>

Datasets for Data Science and Data Mining

<https://www.kdnuggets.com/datasets/index.html>

Federal Reserve Economic Data

<https://fred.stlouisfed.org/>

Repository of Big Datasets

<http://www.datasciencecentral.com/profiles/blogs/big-data-sets-available-for-free>

2. Code Development

To implement the modeling and analysis, it will be necessary to develop code.

The code should be in PySpark. In specific circumstances, Python code may be admissible, but you must discuss this with the instructor first. If small parts of the code are in Python (e.g., reading into a pandas dataframe), this is fine. No other coding languages are admissible.

The code needs to be clearly written and documented in Jupyter notebooks (ipynb format). ***Please clearly describe what the code does at the top of each file. Additionally, place the code's "task" in the filename.***

For full credit (5 PTS), the code needs to include these sections in a clean, commented, and comprehensive manner:

1. Data import and preprocessing (1 PT)

Preprocessing include such tasks as imputing, binning, filtering, outlier treatment, feature engineering,

text processing

2. Data splitting / sampling (0.5 PT)

Sampling may not be needed, but splitting is a must

3. Exploratory data analysis, with at least 2 graphs (1 PT)

4. Model construction, with at least 3 models (1.5 PTS)

Ideally the models are constructed using pipelines. Include a benchmark model (see below).

5. Model evaluation (1 PT)

This should include computation of relevant metrics, and a comparison between models

Note about Modeling using Machine Learning

a. A benchmark model is relatively simple. This could be a regression model with a small number of features (possibly a single feature). This provides a basis for comparison and a sanity check.

b. Include two relatively more sophisticated models (e.g., random forest, gradient boosted tree).

The best model found in your experiments is called the **champion** model.

The model construction process should follow the best practices covered in class, including:

a. Data preprocessing. The required steps will depend on the model, and could include:

- i. Dummy variable construction
- ii. Feature scaling
- iii. Handling missing values and outliers
- iv. Handling semi-structured / unstructured data
- v. Dimensionality reduction (e.g., PCA)

b. Data splitting (train/validation/test sets, for example). The test set should be left out for evaluation purposes. It should NOT be used in training.

c. K-fold cross validation of hyperparameters

Note about Model Evaluation

For all appropriate models (benchmark, champion, and other relevant models), the following should be conducted:

a. Evaluate relevant metrics

For regression, this would include

- i. R-squared

For classification, this would include:

- i. accuracy
- ii. precision, recall, F1 score

- ii. confusion matrix
- iv. area under ROC curve (AUROC)

Depending on the application, additional evaluation could make sense such as lift charts

b. Sensitivity analysis

For your champion model, show relevant metrics for different hyperparameter values. This gives an idea of the model sensitivity.

3. Project Presentation

One of the exciting things about being a data scientist is that they can drive major change at organizations. As a consequence, they can be called upon to communicate with executives. Strong communication skills (to a technical and non-technical audience) is critical.

In the final week of the course, each team will give a group presentation to the class. Each member must present a portion of the project.

Components of the presentation should include:

- i. Executive summary: discuss the research question and what you have found
- ii. Data summary: explain the target (response) variable, the predictors, sampling, etc.
- iii. Variable transformations and preprocessing
- iv. Models constructed (or planned to be constructed)
- v. Model performance
- vi. Conclusions and future research

A presentation earning full points will be strong in:

- i. Content
- ii. Organization / aesthetics
- iii. Delivery

4. Paper

The paper should include the sections below. It could make sense to divide the section writing among teammates; in that case, give the paper a final review for consistency. **The paper should be no more than 6 pages single-spaced.** You can include tables and other artifacts in an appendix; these do not count toward the page limit.

Sections:

i. Abstract

Although the abstract appears first, it should be written last. This includes a quick introduction, an overview of what was done, and a summary of findings.

ii. Data and Methods

iii. Results

iv. Conclusions. The conclusions section can include future work, if there was more time.

- v. Contributions: List the contributions from each team member.

Additional Notes

Project GitHub Repo (recommended but not graded)

Each team member should put their project on their GitHub page.
This should minimally consist of:

- i. *README.md* page that summarizes the project: Purpose, major functionality, class methods
- ii. Organized code
- iii. A *requirements.txt* file listing the required packages

Team and Teammate Evaluation

Each team member needs to make a substantial contribution and needs to be accountable.
If a teammate issue cannot be resolved within the group, please notify the instructor. Students not contributing meaningfully to the project will not receive an A in the course.

Data and Modeling Recommendations

- If your dataset is larger than 50 GB, consider taking a meaningful sample to get within size requirement.
- Early on, drop fields and records that are not needed.
- Categorical variables with many levels can often be bucketed effectively. Conduct EDA to understand how best to bucket.
- Be sure to scale features in regression models and k-means clustering
- When testing if the pipeline works properly, test on a sample of data to save runtime
- When building tree models like GBT, start small to ensure things work. This means small number of trees, shallow depth (e.g., 3). GBT and RF can take a long time to train and are more likely to use up memory.
- When the labels are imbalanced, first split the data into train/test. Next, can change the label proportions in the train set ONLY.

Recommended and Non-Recommended Project Types

These projects tend to go well:

- Supervised learning with text and/or quantitative data; classification, regression
- Sentiment analysis
- Recommendation algorithms

We advise against these types of projects:

- Computer vision, as the required data is massive, the hyperparameter space can be large, and there isn't a variety of data
- Time series modeling, as this requires specific knowledge to do well and there isn't a lot of core Spark support. You can try this but user beware.

Final Notes and Advice

1. If any issues come up during the course of the project, please reach out so we can address them
2. I encourage you to take on a challenging project, but if it cannot be completed in full and on time, select something simpler
3. Practice the presentation beforehand
4. Collaborate with others
5. Meet face-to-face with your teammates
6. Have fun!